

Discrete Random Variables Exercises

Mathematics for AI

Setup

As we discussed, the sample space for a six-sided die is $\Omega_1 = \{1, 2, 3, 4, 5, 6\}$. The probability of throwing each roll is $P(n) = \frac{1}{6}$ for $n \in \Omega_1$.

Now, if we throw two fair six-sided dice the sample space is:

$$\Omega = \Omega_1 \times \Omega_1 = \begin{pmatrix} 1,1 & 1,2 & 1,3 & 1,4 & 1,5 & 1,6 \\ 2,1 & 2,2 & 2,3 & 2,4 & 2,5 & 2,6 \\ 3,1 & 3,2 & 3,3 & 3,4 & 3,5 & 3,6 \\ 4,1 & 4,2 & 4,3 & 4,4 & 4,5 & 4,6 \\ 5,1 & 5,2 & 5,3 & 5,4 & 5,5 & 5,6 \\ 6,1 & 6,2 & 6,3 & 6,4 & 6,5 & 6,6 \end{pmatrix}$$

The probability of throwing any combination of two rolls is: $P(m, n) = \frac{1}{36}$.

Question 1. Sample space of the total

If we throw 2 dice and count the total, what is the sample space of the resultant random variable X ?

$$\Omega_X = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

or in a compact form: $\Omega_X = \{k \in \mathbb{N} : 2 \leq k \leq 12\}$, where $\mathbb{N} = \{0, 1, 2, \dots\}$.

Question 2. Calculating $P(X = x)$

Calculate $P(X = x)$ for each of the elements of the sample space.

$$\begin{array}{ll} P(X = 2) = \frac{1}{36} & P(X = 8) = \frac{5}{36} \\ P(X = 3) = \frac{2}{36} & P(X = 9) = \frac{4}{36} \\ P(X = 4) = \frac{3}{36} & P(X = 10) = \frac{3}{36} \\ P(X = 5) = \frac{4}{36} & P(X = 11) = \frac{2}{36} \\ P(X = 6) = \frac{5}{36} & P(X = 12) = \frac{1}{36} \\ P(X = 7) = \frac{6}{36} & \end{array}$$

Question 3. Probability mass function and distribution function

Using this, construct the probability mass function and the probability distribution function for X .

Probability mass function

$$f_X(x) = \begin{cases} \frac{1}{36} & x = 2, 12 \\ \frac{2}{36} & x = 3, 11 \\ \frac{3}{36} & x = 4, 10 \\ \frac{4}{36} & x = 5, 9 \\ \frac{5}{36} & x = 6, 8 \\ \frac{6}{36} & x = 7 \\ 0 & \text{otherwise} \end{cases}$$

Distribution function

$$F_X(x) = \begin{cases} 0 & x < 2 \\ \frac{1}{36} & 2 \leq x < 3 \\ \frac{3}{36} & 3 \leq x < 4 \\ \frac{6}{36} & 4 \leq x < 5 \\ \frac{10}{36} & 5 \leq x < 6 \\ \frac{15}{36} & 6 \leq x < 7 \\ \frac{21}{36} & 7 \leq x < 8 \\ \frac{26}{36} & 8 \leq x < 9 \\ \frac{30}{36} & 9 \leq x < 10 \\ \frac{33}{36} & 10 \leq x < 11 \\ \frac{35}{36} & 11 \leq x < 12 \\ 1 & x \geq 12 \end{cases}$$

Question 4. Bernoulli distribution

Let X be a random variable with the Bernoulli distribution with parameter p . Calculate the following values.

(a) $\mathbb{E}[X]$

The expectation of a discrete random variable with mass function f is defined as:

$$\mathbb{E}[X] = \sum_{x: f(x) > 0} x f(x)$$

The probability mass function (pmf) of the Bernoulli distribution is:

$$f_X(x) = \begin{cases} p & x = 1 \\ 1 - p & x = 0 \\ 0 & \text{otherwise} \end{cases}$$

so then $\mathbb{E}[X] = (1 \cdot p) + (0 \cdot (1 - p)) = p$

(b) $\mathbb{E}[X^2]$

Lemma: Let X be a random variable, with $g : \mathbb{R} \rightarrow \mathbb{R}$. Then we have that:

$$\mathbb{E}(g(X)) = \sum_x g(x)f_X(x)$$

so:

$$\begin{aligned}\mathbb{E}[X] (X^2) &= \sum_{x \in \{0,1\}} x^2 f_X(x) \\ &= (0)^2 \cdot (1-p) + (1)^2 \cdot p \\ &= 0 + p \\ &= p\end{aligned}$$

(c) $\text{Var}(X)$

The variance of a random variable X is defined as:

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$$

and from this Lemma:

Lemma: Let X be a random variable. Then

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

So for this case:

$$\begin{aligned}\text{Var}(X) &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \\ &= p - p^2 \\ &= p(1-p)\end{aligned}$$

Question 5. Poisson: chocolate chips in a cookie

Assuming the number of chocolate chips in a cookie has a Poisson distribution, if we want the probability that a randomly chosen cookie has at least two chocolate chips to be greater than 0.99, what is the smallest integer value of the mean of the distribution that ensures this probability?

Set-up.

(a) **Poisson: chocolate chips in a cookie**

Take X as the number of chocolate chips in a random cookie and X has a $\text{Poisson}(\lambda)$ distribution. The pmf is then:

$$f_X(x) = \frac{\lambda^x}{x!} e^{-\lambda}, \quad x \in \{0, 1, 2, 3, \dots\}$$

We want $P(X \geq 2) > 0.99$. Trying to calculate this directly would require an infinite sum. But there is an alternative, we can use the complement of ≥ 2 which is finite :

$$P(X \geq 2) = 1 - P(X = 0) - P(X = 1)$$

From the pmf stated above:

$$P(X = 0) = \frac{\lambda^0}{0!} e^{-\lambda} = e^{-\lambda}$$

$$P(X = 1) = \frac{\lambda^1}{1!} e^{-\lambda} = \lambda e^{-\lambda}$$

We can then substitute these values into the probability condition:

$$1 - e^{-\lambda} - \lambda e^{-\lambda} > 0.99$$

$$e^{-\lambda}(1 + \lambda) < 0.01$$

There is no clear one solution, we need to then test different values of λ till the condition holds:

λ	$e^{-\lambda}(1 + \lambda)$
1	0.7357..
4	0.0915...
6	0.0173...
7	0.0072...
8	0.0028...

Answer: So based on the different levels of λ tested above the minimum integer values of the mean of the distribution λ is $\lambda = 7$